

Field Visit and Environmental Issue Report

Objective

The objective of this field visit was to observe and analyze environmental conditions at a protected biodiversity site, with special focus on identifying ecological significance, environmental challenges, human impact, and conservation measures. The visit aimed to understand the importance of wetland ecosystems and the role they play in maintaining ecological balance.

Site Visited:

Thane Creek Flamingo Sanctuary, Maharashtra

Date of Visit:

20/11/2025

Observations

Environmental Issue(s) Observed

Thane Creek Flamingo Sanctuary is a coastal wetland ecosystem known for its rich biodiversity, especially migratory flamingos. During the visit, the sanctuary showed both **healthy ecological features** and **environmental concerns**. Large populations of flamingos were observed feeding in shallow waters. Mangrove vegetation was widespread, indicating the ecological importance of the area.

However, signs of environmental stress were also evident, such as water pollution, floating plastic waste, and nearby industrial activity affecting the creek.

Sources and Impacts

The major sources impacting the sanctuary include:

- Industrial discharge from nearby industrial zones
- Domestic sewage entering the creek through drainage channels
- Plastic and solid waste carried by tidal movement
- Noise and air pollution from nearby transport routes

These factors negatively affect water quality and disturb the natural habitat of birds and aquatic organisms. Polluted water can reduce food availability for flamingos, particularly algae and small aquatic organisms.

Community or Ecological Effects

The environmental conditions of Thane Creek have significant ecological implications:

- Threat to migratory birds due to habitat degradation
- Risk to mangrove ecosystems which act as natural buffers
- Decline in aquatic organisms affecting the food chain
- Reduced eco-tourism potential if pollution increases

Despite these challenges, conservation efforts have helped maintain the sanctuary as a critical winter habitat for thousands of flamingos.

Data Collected

Photos

Photographs were taken to document:

- Flamingos feeding in the creek
- Mangrove vegetation
- Areas affected by waste accumulation
- Surrounding industrial landscape

Interviews

Informal interaction with local guides and visitors revealed:

- Increase in flamingo population in recent years due to protection efforts
- Ongoing concerns about pollution and waste management
- Importance of public awareness for conservation success

Sample Readings (if applicable)

No laboratory testing was conducted. However, visual observations included:

- Slight discoloration of water in certain areas
- Presence of floating debris
- Clear distinction between protected and polluted zones

Recommendations

Based on the field observations, the following recommendations are suggested:

- Strict control and monitoring of industrial effluent discharge
- Regular cleaning of creek waters and surrounding areas
- Strengthening mangrove conservation and plantation programs
- Awareness campaigns for visitors and nearby communities
- Promoting eco-friendly tourism practices
- Continuous biodiversity monitoring by environmental authorities

Acknowledgement

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Prepared by:

1. **Afan Pawaskar - FY-BSc-IT**
(Roll no.44)
2. **Sameer Mohammed Amin Shaikh - FY-BSc-IT**
(Roll no.93)
3. **Dalvi Tanzeel Ahmed- FY-BSc-IT**
(Roll no.30)